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EEEE3054

Renewable Generation Technologies

Safety Standards, Guidance and Regulation

Introduction

- All PV and related products must be manufactured in accordance with safety and performance regulations. Manufacturers and end users treat product certification as the reference of trust.
- A number of codes and standards have been developed for safe and reliable development of PV power systems.
- It should be noted that safety standards are different from country to country.

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CERTIFICATION OF PV MODULES

CERTIFICATION OF PV MODULES

Introduction

- The common safety standards of PV module products include IEC-61215, IEC-61646, IEC-61730, and UL-1703.
- As well as the PV cells, other components used for PV module construction should be evaluated to ensure their safety, reliability, quality, and performance.
- These include polymeric materials, junction boxes, connectors, and connecting cables.

CERTIFICATION OF PV MODULES

Standards

Standards	Title
IEC 61215	Crystalline silicon terrestrial photovoltaic (PV) modules – Design qualification and type approval
IEC 61646	Thin-film terrestrial photovoltaic (PV) modules– Design qualification and type approval
IEC 61730	Photovoltaic (PV) module safety qualification: Part 1– Requirements for construction Part 2– Requirements for testing
UL 1703	Standard for flat-plate photovoltaic modules and panels
UL 2703	Standard for mounting systems, mounting devices, clamping/ retention devices, and ground lugs for use with flat-plate photovoltaic modules and panels
UL 3703	Standard for solar trackers for use with distributed energy resources

CERTIFICATION OF PV MODULES

Regulation Bodies

- The International Electrotechnical Commission (IEC) is a not-for-profit, non-governmental organization, which publishes consensus-based standards and manages conformity assessment systems for electrical and electronic products.
- In Europe, the leading organizations for certification are the Germany-based TÜV Rheinland and VDE, the Association for Electrical, Electronic, and Information Technologies.
- In North America, the UL-1703 standard is commonly applied and PV modules are validated by either the Underwriters Laboratories (UL) or the Electrical Testing Laboratory (ETL), a division of the Intertek Group.

CERTIFICATION OF PV MODULES

Certification

- For certification, PV module manufacturers must submit the product specification, full material lists, product samples, and related documents to the certification agencies.
- After an initial evaluation, the certification body starts lab tests to assess construction and performance.
- This comprehensive evaluation guarantees the product's performance, quality, and safety. When all tests have been passed, the certification can be issued.
- The end users can search for certified products through the online databases provided by the certification agencies.

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INTERCONNECTION STANDARDS

INTERCONNECTION STANDARDS

Introduction

- PV converters must meet safety standards of UL 1741 and IEC 62109 for grid interconnection.
- UL 1741 was first introduced in the late 1990s and revised in 2001. It is the product safety standard that defines the testing and certification requirements for inverters, converters, controllers, and other interconnection system equipment.
- In most jurisdictions of the USA and Canada, compliance with UL 1741 is mandatory for any PV power system that will be connected to the utility grid.
- The performance testing of UL 1741 is comprehensive, which will not be covered here.

INTERCONNECTION STANDARDS

Standards

Standards	Title
UL1741	Standard for Inverters, Converters, Controllers and Interconnection System Equipment for Use With Distributed Energy Resources
IEC 62109-1:2010	Safety of power converters for use in photovoltaic power systems–Part 1: General requirements
IEC 62109-2:2011	Safety of power converters for use in photovoltaic power systems–Part 2: Particular requirements for inverters

Anti-Islanding

- In grid-connected PV systems, the term of islanding refers to the condition in which the distributed resource (DR) continues to supply electric power even though the electric utility is malfunctioning.
- Islanding involves a potential of safety risk to utility workers.
- Therefore, the function of anti-islanding or islanding protection is mandatory in UL 1741 against the continuous power supply from the inverter, which is isolated from the remainder of the electric utility system.
- To meet the requirement, research has been conducted to detect the grid failure accurately and efficiently regardless of the presence of harmonics and noise.
- Anti-islanding includes both passive and active methods. The active anti-islanding algorithm is more effective than passive counterparts to avoid a non-detection zone, but the conventional algorithm introduces disturbance to the utility grid.

INTERCONNECTION STANDARDS

Anti-Islanding

- The PV inverters need to disconnect from the grid in case of abnormal grid conditions in terms of voltage and frequency.
- This response is to ensure the safety of utility maintenance personnel and the general public, as well as to avoid damage to connected equipment, including the PV system.
- This requirement is commonly referred to as “anti-islanding protection.”

Voltage Deviations

- All discussions regarding system voltage refer to the local nominal voltage.
- The voltages in RMS are measured at the point of utility connection.
- The disconnection time refers to the time between the abnormal condition occurring and the inverter ceasing to energize the utility line.
- The inverter controls shall actually remain connected to the utility to allow sensing of utility electrical conditions for use by the ‘reconnect’ feature.
- The purpose of the allowed time delay is to ride through short-term disturbances to avoid excessive nuisance tripping.

Disconnection Time for Voltage Variations

IEEE 1547		IEC 61727		VDE0126-1-1	
Voltage range (%)	Disconnection time (s)	Voltage range (%)	Disconnection time (s)	Voltage range (%)	Disconnection time (s)
$V < 50$	0.16	$V < 50$	0.10	$110 \leq V < 85$	0.2
$50 \leq V < 88$	2.00	$50 \leq V < 85$	2.00		
$110 \leq V < 120$	1.00	$110 \leq V < 135$	2.00		
$V \geq 120$	0.16	$V \geq 135$	0.05		

Frequency Deviations

- The purpose of the allowed range and time delay is to ride through short-term disturbances to avoid excessive nuisance tripping in weak-grid situations.

IEEE 1547		IEC 61727		VDE0126-1-1	
Frequency (Hz)	Disconnection time (s)	Frequency (Hz)	Disconnection time (s)	Frequency (Hz)	Disconnection time (s)
$59.3 < f < 60.5$*	0.16	$f_n - 1 < f < f_n + 1$	0.2	$47.5 \leq f < 50.2$	0.2

* For IEEE 1547-2018, 57 Hz < f < 61.8 Hz

Note: While IEC 61727 sets these general guidelines, specific local grid codes (such as EN50438, VDE, or local TNB/MEA regulations) may adopt slightly tighter or different tolerance values (e.g., 50Hz ±2% or **±0.5 Hz**) for the final interconnection agreement.

INTERCONNECTION STANDARDS

Reconnection after Trip

- After a disconnection caused by abnormal voltage or frequency conditions the inverter can be reconnected only in the conditions.

IEEE 1547	IEC 61727	VDE0126-1-1
88 < V < 110 (%) **AND** 59.3 < f < 60.5 (Hz)	88 < V < 110 (%) **AND** $f_n - 1 < f < f_n + 1$ (Hz) **AND** Minimum delay of 3 minutes	

INTERCONNECTION STANDARDS

Power Quality

- The quality of the power provided by the photovoltaic system for the local AC loads and for the power delivered to the utility is governed by practices and standards on voltage, flicker, frequency, harmonics and power factor.
- Deviation from these standards represents out-of-bounds conditions and may require disconnection of the photovoltaic system from the utility.

DC Current Injection

- DC current injection in the utility can saturate the distribution transformers, leading to over heating and trips.
- For the conventional PV systems with galvanic isolation, this problem is minimized, but with the new generation of transformerless PV inverters increased attention is required in this matter.

DC Current Injection

- For IEEE 1574 and IEC 61727 the DC component of the current should be measured by using harmonic analysis (FFT) and there is no maximum trip time condition.
- During the test the measured DC component should be below the limits for different loading conditions (1/3, 2/3 and 3/3 of the nominal load).
- For VDE 0126-1-1 this condition requires a special designed current sensor that can detect this threshold and disconnect within the required trip time.

DC current injection limitation

IEEE 1547	IEC 61727	VDE0126-1-1
I_{DC} < 0.5 % I_{rated}	I_{DC} < 1 % I_{rated}	I_{DC} < 1 A Maximum trip time 0.2 s

Current Harmonics

- The PV system output should have low current-distortion levels to ensure that no adverse effects are caused to other equipment connected to the utility system.

Maximum current harmonics in odd order

IEEE 1547 and IEC 61727						
Individual harmonic order (odd) (%)	h <11	11≤ h <17	17≤ h < 23	23≤ h <35	h ≤ 35	Total harmonic distortion THD (%)
	4.0	2.0	1.5	0.6	0.3	5.0



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QUESTIONS?